

Python Tip: `pathlib`

Need to work with file paths in Python?

`pathlib` makes paths easy to find, build, inspect, and change without manually joining or slicing strings!



Create a Path

Import `Path` from `pathlib`. You can get useful folders from Python or turn your own path string into a `Path` object.

```
from pathlib import Path

current_folder = Path.cwd()
home_folder = Path.home()
file_path = Path("reports/2026/sales.csv")
```

`Path.cwd()` gets the current working folder. `Path.home()` gets the current user's home folder. Neither command creates or opens anything.

Build Paths with /

`pathlib` makes it extremely easy to build paths. Use the `/` operator to join folders and file names.

```
from pathlib import Path

reports = Path("reports")
file_path = reports / "2026" / "sales.csv"

print(file_path)
```

```
reports/2026/sales.csv
```

There is no need to add slashes to strings by hand. `pathlib` joins the pieces for us.

Inspect a Path

A `Path` object makes it easy to get the most useful parts of a file path.

```
from pathlib import Path

file_path = Path("reports/2026/sales.csv")

print("Name:", file_path.name)
print("Stem:", file_path.stem)
print("Extension:", file_path.suffix)
print("Folder:", file_path.parent)
```

```
Name: sales.csv
Stem: sales
Extension: .csv
Folder: reports/2026
```

The `stem` is the name without the extension. The `suffix` is the extension, including the dot.

Change a Path

You can create a path with a different file name or extension without editing the string yourself.

```
from pathlib import Path

csv_path = Path("reports/sales.csv")

json_path = csv_path.with_suffix(".json")
old_path = csv_path.with_name("sales-old.csv")

print("New type:", json_path)
print("New name:", old_path)
```

```
New type: reports/sales.json
New name: reports/sales-old.csv
```

These methods return new `Path` objects. They do not rename, move, or change any files.

Paths Depend on the OS

`Path` follows the rules of the operating system running your code. On macOS and Linux, it creates a `PosixPath`. On Windows, it creates a `WindowsPath`.

- **POSIX:** `/reports/2026/sales.csv`
- **Windows:** `C:\reports\2026\sales.csv`

This automatic behavior is useful for paths your program creates and uses on the current computer. To inspect path text written for a specific system, use `PureWindowsPath` or `PurePosixPath`.

Wrap-Up

Use `Path` to build and inspect file paths without manually changing strings.

Use `/` to build paths and properties like `name`, `stem`, `suffix`, and `parent` to inspect them.

Remember that `Path` follows the current operating system's rules. Use `PureWindowsPath` or `PurePosixPath` for path text from a specific system.

Follow me for more tips.

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